Class11

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Background

In this lab, we analyze AlphaFold structure prediction results. We will read the predicted PDB models, compare them with RMSD, examine confidence scores with pLDDT, analyze model flexibility with RMSF, inspect PAE scores, and look at sequence conservation.

```
library(bio3d)
library(pheatmap)
library(jsonlite)
```

Read AlphaFold results

First, unzip the AlphaFold results file.

```
unzip("test_94b5b.result.zip", exdir = "test_94b5b")
```

Make a vector of input PDB file names that we can read into R.

```
results_dir <- "test_94b5b"

pdb_files <- list.files(
  path = results_dir,
  pattern = "\\.pdb$",
  full.names = TRUE
)

basename(pdb_files)
```

```
[1] "test_94b5b_unrelaxed_rank_001_alphafold2_ptm_model_4_seed_000.pdb"
[2] "test_94b5b_unrelaxed_rank_002_alphafold2_ptm_model_5_seed_000.pdb"
[3] "test_94b5b_unrelaxed_rank_003_alphafold2_ptm_model_3_seed_000.pdb"
[4] "test_94b5b_unrelaxed_rank_004_alphafold2_ptm_model_1_seed_000.pdb"
[5] "test_94b5b_unrelaxed_rank_005_alphafold2_ptm_model_2_seed_000.pdb"
```

Read all PDB models and superpose them.

```
pdbs <- pdbaln(pdb_files, fit = TRUE, exefile = "msa")
```

```
Reading PDB files:
test_94b5b/test_94b5b_unrelaxed_rank_001_alphafold2_ptm_model_4_seed_000.pdb
test_94b5b/test_94b5b_unrelaxed_rank_002_alphafold2_ptm_model_5_seed_000.pdb
test_94b5b/test_94b5b_unrelaxed_rank_003_alphafold2_ptm_model_3_seed_000.pdb
test_94b5b/test_94b5b_unrelaxed_rank_004_alphafold2_ptm_model_1_seed_000.pdb
test_94b5b/test_94b5b_unrelaxed_rank_005_alphafold2_ptm_model_2_seed_000.pdb
.....

Extracting sequences

pdb/seq: 1   name: test_94b5b/test_94b5b_unrelaxed_rank_001_alphafold2_ptm_model_4_seed_000.
pdb/seq: 2   name: test_94b5b/test_94b5b_unrelaxed_rank_002_alphafold2_ptm_model_5_seed_000.
pdb/seq: 3   name: test_94b5b/test_94b5b_unrelaxed_rank_003_alphafold2_ptm_model_3_seed_000.
```

```
pdb/seq: 4   name: test_94b5b/test_94b5b_unrelaxed_rank_004_alphafold2_ptm_model_1_seed_000.p
pdb/seq: 5   name: test_94b5b/test_94b5b_unrelaxed_rank_005_alphafold2_ptm_model_2_seed_000.p
```

```r
pdbs
```

```
                                1        .         .         .         .         50
[Truncated_Name:1]test_94b5b   PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:2]test_94b5b   PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:3]test_94b5b   PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:4]test_94b5b   PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:5]test_94b5b   PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
                               **************************************************
                                1        .         .         .         .         50

                               51        .         .         .         .        99
[Truncated_Name:1]test_94b5b   GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:2]test_94b5b   GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:3]test_94b5b   GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:4]test_94b5b   GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:5]test_94b5b   GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
                               *************************************************
                               51        .         .         .         .        99

Call:
  pdbaln(files = pdb_files, fit = TRUE, exefile = "msa")

Class:
  pdbs, fasta

Alignment dimensions:
  5 sequence rows; 99 position columns (99 non-gap, 0 gap)

+ attr: xyz, resno, b, chain, id, ali, resid, sse, call
```

RMSD analysis

RMSD measures structural distance between coordinate sets. We can calculate RMSD between all pairs of AlphaFold models.

```r
rd <- rmsd(pdbs, fit = TRUE)
```

```
Warning in rmsd(pdbs, fit = TRUE): No indices provided, using the 99 non NA positions
```

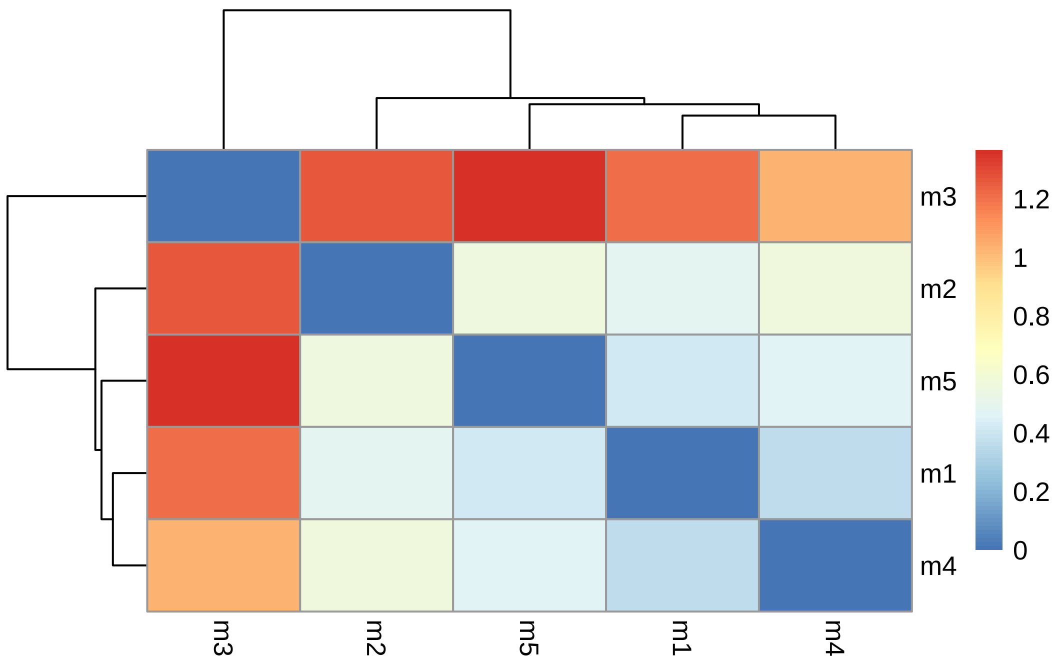
```r
range(rd)
```

```
[1] 0.000 1.367
```

Make RMSD heatmap

```r
colnames(rd) <- paste0("m", 1:ncol(rd))
rownames(rd) <- paste0("m", 1:nrow(rd))

pheatmap(rd)
```

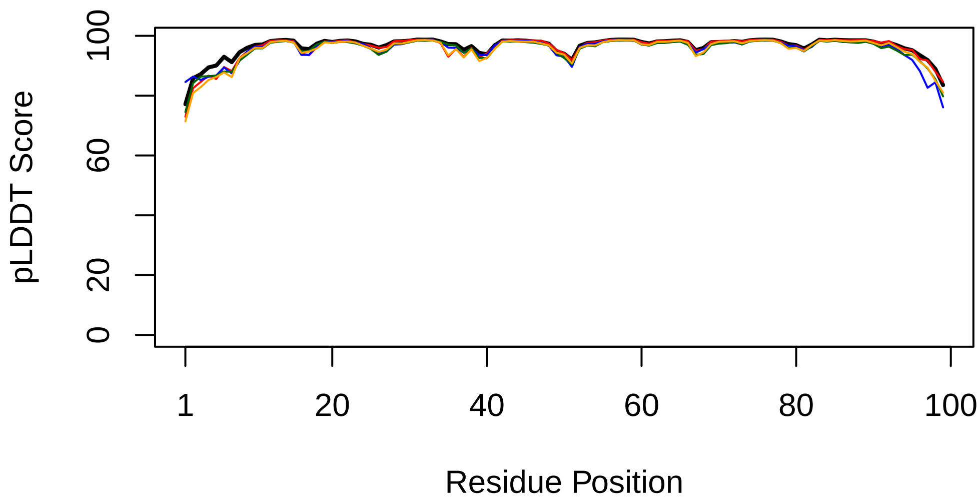


Plot pLDDT scores

The AlphaFold PDB files store pLDDT confidence scores in the B-factor column. Higher pLDDT means higher confidence.

```r
plotb3(
  pdbs$b[1,],
  typ = "l",
  lwd = 2,
  ylab = "pLDDT Score",
  xlab = "Residue Position"
)

points(pdbs$b[2,], typ = "l", col = "red")
points(pdbs$b[3,], typ = "l", col = "blue")
points(pdbs$b[4,], typ = "l", col = "darkgreen")
points(pdbs$b[5,], typ = "l", col = "orange")
```



Core fitting

Next, find the most consistent rigid core of the models and use this core for fitting.

```r
core <- core.find(pdbs)
```

```
core size 98 of 99  vol = 4.608
core size 97 of 99  vol = 3.729
```

```
 core size 96 of 99  vol = 2.981
 core size 95 of 99  vol = 2.312
 core size 94 of 99  vol = 1.831
 core size 93 of 99  vol = 1.359
 core size 92 of 99  vol = 0.992
 core size 91 of 99  vol = 0.606
 core size 90 of 99  vol = 0.38
 FINISHED: Min vol ( 0.5 ) reached
```

```
core.inds <- print(core, vol = 0.5)
```

```
# 91 positions (cumulative volume <= 0.5 Angstrom^3)
  start end length
1     3   3      1
2     7  96     90
```

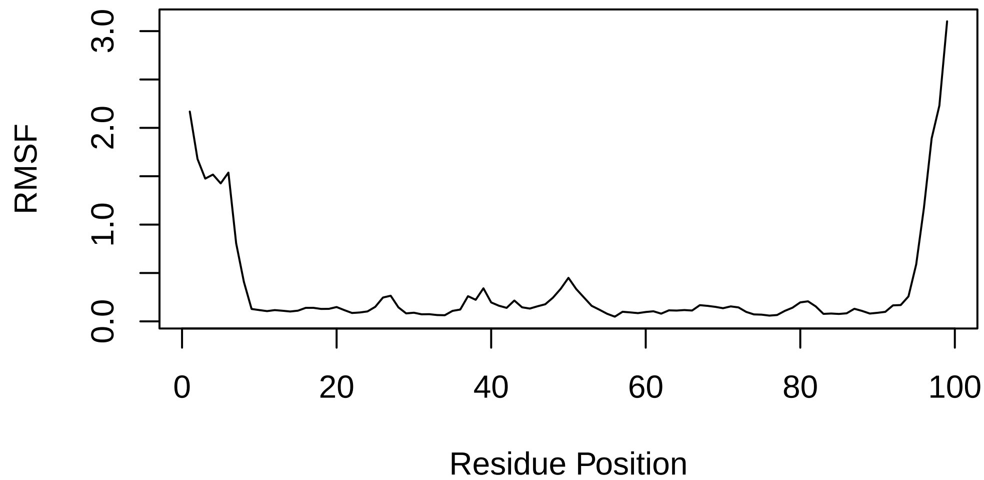
```
xyz <- pdbfit(
  pdbs,
  core.inds,
  outpath = "corefit_structures"
)
```

RMSF analysis

RMSF measures conformational variation across residue positions.

```
rf <- rmsf(xyz)

plot(
  rf,
  type = "l",
  ylab = "RMSF",
  xlab = "Residue Position"
)
```



PAE analysis

AlphaFold also produces Predicted Aligned Error, or PAE, values in JSON files.

```
pae_files <- list.files(
  path = results_dir,
  pattern = ".*model.*\\.json",
  full.names = TRUE
)

pae_files
```

```
[1] "test_94b5b/test_94b5b_scores_rank_001_alphafold2_ptm_model_4_seed_000.json"
[2] "test_94b5b/test_94b5b_scores_rank_002_alphafold2_ptm_model_5_seed_000.json"
[3] "test_94b5b/test_94b5b_scores_rank_003_alphafold2_ptm_model_3_seed_000.json"
[4] "test_94b5b/test_94b5b_scores_rank_004_alphafold2_ptm_model_1_seed_000.json"
[5] "test_94b5b/test_94b5b_scores_rank_005_alphafold2_ptm_model_2_seed_000.json"
```

Read the first and last model PAE files.

```
pae1 <- read_json(pae_files[1], simplifyVector = TRUE)
pae_last <- read_json(pae_files[length(pae_files)], simplifyVector = TRUE)

pae1$max_pae
```

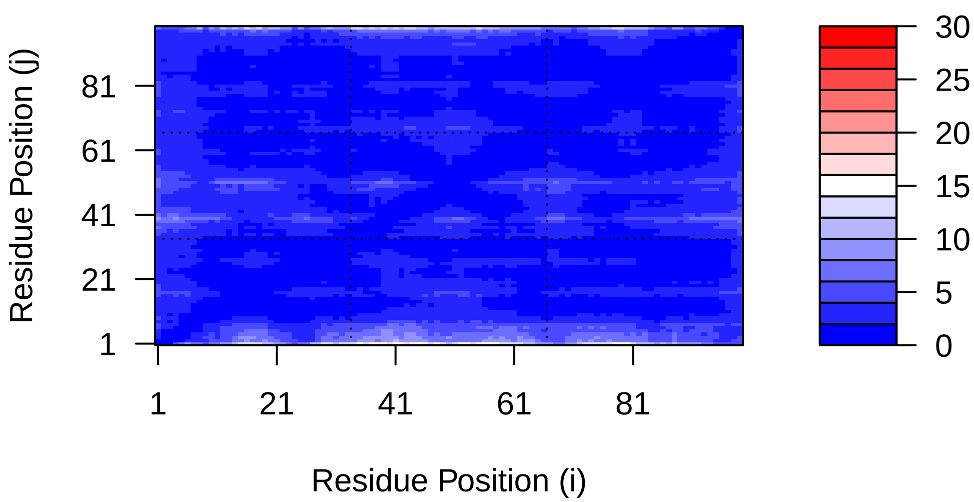
```
[1] 16.96875
```

```
pae_last$max_pae
```

```
[1] 19.3125
```

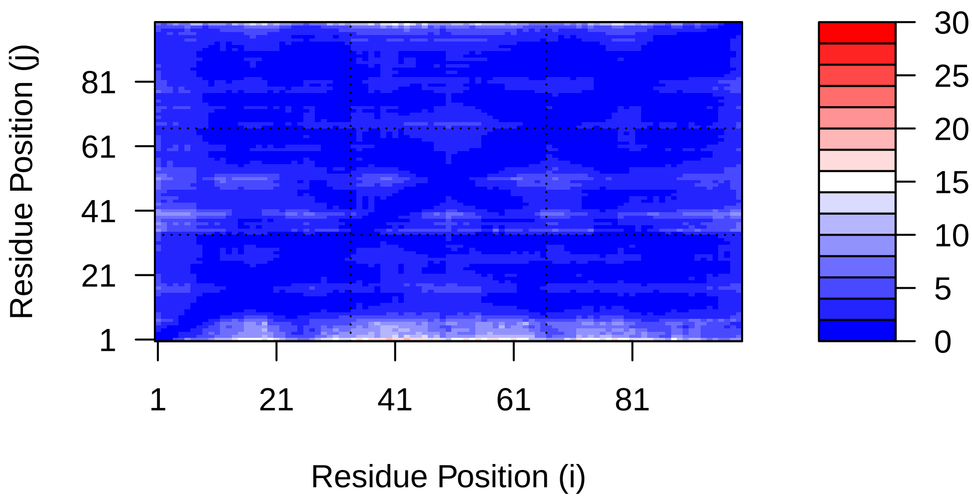
PAE plot for model 1

```
plot.dmat(
  pae1$pae,
  xlab = "Residue Position (i)",
  ylab = "Residue Position (j)",
  grid.col = "black",
  zlim = c(0, 30)
)
```



PAE plot for model 5

```
plot.dmat(
  pae_last$pae,
  xlab = "Residue Position (i)",
  ylab = "Residue Position (j)",
  grid.col = "black",
  zlim = c(0, 30)
)
```



Conservation analysis

```
aln_file <- list.files(
  path = results_dir,
  pattern = "\\.a3m$",
  full.names = TRUE
)

aln_file
```

```
[1] "test_94b5b/test_94b5b.a3m"
```

Read alignment

```
aln <- read.fasta(aln_file[1], to.upper = TRUE)
```

```
[1] " ** Duplicated sequence id's: 101 **"
```

```
dim(aln$ali)
```

```
[1] 5397  132
```

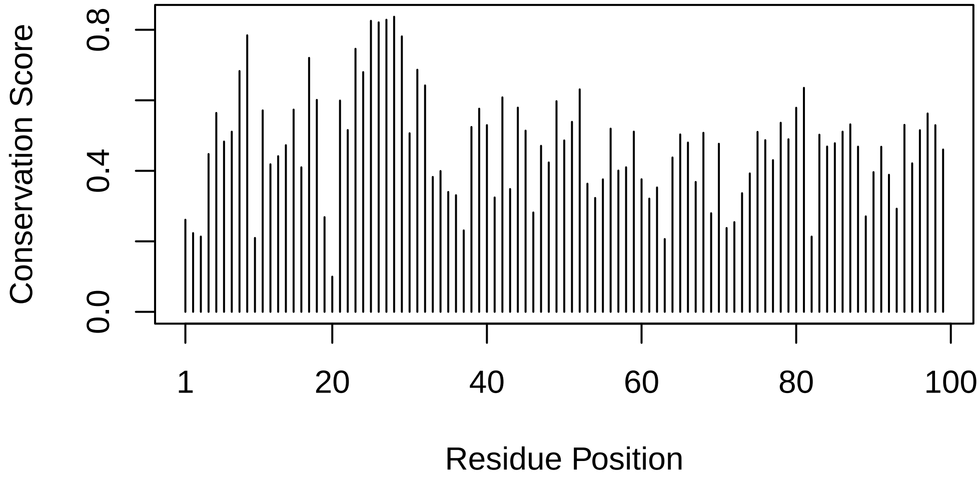
Score residue conservation.

```
sim <- conserv(aln)

m1.pdb <- read.pdb(pdb_files[1])

nres <- length(unique(m1.pdb$atom$resno))

plotb3(
  sim[1:nres],
  ylab = "Conservation Score",
  xlab = "Residue Position"
)
```



Consensus sequence

```r
con <- consensus(aln, cutoff = 0.9)

con$seq
```

```
  [1] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [19] "-" "-" "-" "-" "-" "-" "D" "T" "G" "A" "-" "-" "-" "-" "-" "-" "-" "-"
 [37] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [55] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [73] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [91] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[109] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[127] "-" "-" "-" "-" "-" "-"
```

Save conservation scores into occupancy column of model 1

```
occ <- vec2resno(
  sim[1:nres],
  m1.pdb$atom$resno
)

write.pdb(
  m1.pdb,
  o = occ,
  file = "m1_conserv.pdb"
)
```

Conclusion

The AlphaFold models were compared using RMSD, pLDDT, RMSF, PAE, and conservation analysis. The RMSD heatmap shows how similar or different the predicted models are from each other. The pLDDT plot shows confidence across residue positions. The RMSF plot highlights regions that vary more between models. The PAE plots show prediction error across residue pairs. The conservation analysis identifies more conserved positions in the sequence alignment.